3 How to be a physicalist

Chapter overview

- Explains the core claims of behaviourism, the mind-brain identity theory, and functionalism
- Considers eliminativism as an approach to mentality
- Examines the respective roles of philosophical analysis and scientific inquiry in understanding the mind.

In the previous chapter, we identified physicalism as a key position in the philosophy of mind. According to the physicalist, mental phenomena are 'nothing over and above' physical phenomena. We saw also that one way of explicating that slogan is in terms of metaphysical supervenience, together with the idea that the idea that the metaphysical supervenience of the mental on the physical is not just a brute, unexplainable, fact but can be rendered intelligible.

The aim of this chapter is to consider four of the most influential approaches to the mind: behaviourism, the identity theory, functionalism, and eliminativism. Although not all of these approaches are strictly speaking committed to physicalism, they are all 'physicalism friendly', for each promises both to secure the dependence of the mental on the physical and to provide an explanation for that dependence. We will consider some of the main motivations for each of these theories and some of the central challenges that they face, but our primary focus here isn't on evaluating these positions but on understanding them.

3.1 Behaviourism

We begin with behaviourism, an approach to the mind that had its heyday between the 1930s and 1960s. As the label suggests, behaviourists take behaviour to be the key to mentality. There are, however, different traditions within behaviourism. In psychology, the most influential versions of behaviourism focused on methodological and explanatory issues. For

DOI:10.4324/9781003225348-4

example, the behaviourists J. B. Watson (1878–1958) and B. F. Skinner (1904–90) held that the study of the mind should focus on the study of an organism's behavioural responses, and the ways in which those responses are altered by its environment and learning history. The psychological behaviourists eschewed reference to inner mental states and processes, arguing that they are not needed for the explanation of human behaviour. Indeed, Watson argued that positing such states wouldn't even be scientifically legitimate.

Although psychological behaviourism has had an influence on the philosophy of mind, the more profound behaviourist influence is due to a view that I will refer to as 'analytical behaviorism'. (It's also known as 'logical behaviourism' and 'philosophical behaviourism'.) Analytical behaviourists held that mental states can be analysed in terms of behavioural dispositions. For example, what it is to believe that it's raining outside just is to be disposed to behave in certain ways – for example, to put one a raincoat before heading outside or nod one's head in response to the question 'Is it raining outside?' The view is so called because its advocates held that an account of the relevant behavioural dispositions is implicit in the meaning of psychological terms, and thus that the behavioural analyses of mental phenomena qualify as analytical truths (claims that are true in virtue of their meaning). Someone who knows what 'pain' means knows that what it is to be in pain is a matter of having (say) the disposition to wince and to move away from a noxious stimulus. Importantly, what counts as behaviour here must be understood in non-psychological terms. Thus, what it is to 'put on one's raincoat' isn't a matter of intentionally putting on something that one believes to be one's raincoat, but is instead to be understood in terms of moving one's limbs so as to make it the case that one is in fact wearing a raincoat.

Behaviourism has a number of motivations, two of which continue to exert a significant influence on the philosophy of mind. The first motivation is epistemic. Our ordinary grounds for ascribing mental states to other people involve close attention to their behaviour – to what they do and say. The fact that Bonnie is smiling provides evidence that she's happy, and the fact that Clyde is yelling is naturally taken to indicate that he's angry. Now, if mental states just are behavioural dispositions (as the analytical behaviourists claimed), then we would have a straightforward account for why we are justified in ascribing happiness to people who smile and anger to people who yell. Whatever the shortcomings of behaviourism, there is clearly much that is attractive in any view of the mental which promises to vindicate our ordinary practice of ascribing mental states to others on the basis of behaviour.

A second, and closely related, motivation for behaviourism is the desire to reject a Cartesian conception of the mental as a private realm that is accessible only from the first-person perspective. In this regard, behaviourism drew significant inspiration from Gilbert Ryle (1900–76) and Ludwig Wittgenstein (1889–1951). Although neither Ryle nor Wittgenstein were full-blooded behaviourists, their accounts of the mind had much in common with behaviourism, and their influence did much to elevate that of behaviourism. Both Ryle and Wittgenstein rejected the dualist conception of the self as an immaterial substance that controls the body – what Ryle's parodied as 'the Ghost in the Machine' in his book *The Concept of Mind* (1949). They also argued that the meaning of our psychological vocabulary cannot be prised apart from its use to describe ordinary activity. As Wittgenstein put it in his *Philosophical Investigations* (1953/2009: §580), 'an "inner processes" stands in need of outward criteria'.

We will shortly consider a number of objections to behaviourism, but let us first reflect on why behaviourism qualifies as a form of physicalism. First, behaviourism satisfies the modal constraint on physicalism, for if mental phenomena are identical to behavioural dispositions then the mental facts are nothing over and above the physical facts. Behaviourism also satisfies the intelligibility constraint, for if mental phenomena are behavioural dispositions then the supervenience of the mental on the physical isn't primitive but can instead be fully explained.

The question, of course, is whether the behaviourist account of mentality is plausible. It certainly has some appeal with respect to certain kinds of mental phenomena. Consider, for example, character traits, such as being shy, curious, or quick-witted. Arguably, all there is to being shy is being disposed to behave in certain kinds of ways in certain contexts. But it is implausible to suppose that mental phenomena in general can be identified with behavioural dispositions in the ways in which analytical behaviourists held. Let us consider three reasons for this.

First, behaviourism struggles to account for the privacy of the mental. As we noted in the first chapter, we seem to enjoy a kind of direct and secure access to certain kinds of mental states. You have a kind of knowledge of your current emotional experiences that I don't (and possibily cannot) share. Similarly, you know what you are currently thinking in ways that I don't. Behaviourism renders the privacy of the mental mysterious, for – in principle at least – a person's behavioural dispositions are accessible from the third-person perspective. Many of those in the behaviourist tradition tend to respond to this challenge by downplaying the privacy of the mental, but few have found that response compelling.

Second, although many mental phenomena are correlated with behavioural responses, this link appears to be neither analytic nor necessary. Consider the relationship between pain and wincing. As it happens, being in pain is correlated with wincing in humans, but it seems perfectly coherent to suppose that there might be creatures for which this correlation fails to hold. Consider super-spartans – a kind of creature that can experience pain, but have none of the behavioural dispositions that we associate with pain (Putnam 1963). Super-spartans not only remain silent when in pain, they aren't even *disposed* to cry or whimper. Behaviourists must rule out the very possibility of super-spartans, but on the face of things super-spartans seem to be perfectly coherent. Equally coherent is the possibility of beings – let's call them 'actors' – who are disposed to produce pain behaviours (i.e., the kind of behaviours that in us are associated with pain) despite the fact that they aren't in pain. But again, if pain is identified with the disposition to produce a range of the behaviours that are associated with pain in us, then it would follow that actors are not a genuine possibility.

A third problem for behaviourism concerns the threat of circularity: to provide a behavioural analysis of one kind of mental state, one must invariably mention other kinds of mental states (Chisholm 1957). I might believe that it's raining outside, but I won't answer the question 'Is it raining outside?' in the affirmative if I desire to hide my weather-related beliefs from you, nor will I take an umbrella with me if I believe that the rain will soon cease and that taking an umbrella is more trouble than it's worth. What one is disposed to do in any particular context depends on one's entire network of beliefs, desires, and intentions, not to mention one's habits and emotional state. Thus, giving a full specification of any one mental state will inevitably involve reference to mental states of other kinds. But if that's right, then

the behaviourist hasn't really provided a full account of mentality, or shown how mental phenomena might be 'nothing over and above' physical phenomena.

3.2 The mind-brain identity theory

Although it had relatively few explicit adherents, analytical behaviourism was extremely influential in the philosophy of mind until the late 1950s and early 1960s, at which point it was usurped by the mind-brain identity theory (also known as 'central-state materialism'). Identity theorists argue that mental states – or at least certain classes of mental states – are identical to brain states. For example, the identity theorist might identity pain with a particular kind of activity in the so-called 'pain matrix', a network involving the primary and secondary somatosensory cortices, the insula, and the anterior cingulate cortex, and which is typically active when people experience transient pain (although see page 168). One can find hints of the identity theory in the work of 17th- and 18th-century materialists such as Thomas Hobbes, Julien Offray de La Mettrie, and Baron d'Holbach, but the detailed development of the view did not occur until the middle decades of the 20th century. Some of its most significant advocates include Herbert Feigl (1958), U. T. Place (1956), J.J.C. Smart (1959), David Armstrong (1968), and David Lewis (1966). Although not as dominant as it once was, the identity theory remains influential (Box: Anomalous monism and token identities).

Box: Anomalous monism and token identities

The dominant version of the identity theory is known as the *type* identity theory, for it holds that types of mental states (i.e., mental properties) are identical to types of neural states (i.e., neural properties). However, there is another version of the identity account – the *token* identity theory – which has also been very influential. The token identity theory is associated most closely with a view of the mind known as *anomalous monism*. Developed by Donald Davidson (1917–2003) in the 1970s, the anomalous monist holds that although there are no laws that can be formulated using both physical and mental predicates (that's the 'anomalous' part), every particular mental event is identical to a particular physical event (that's the 'monism' part). According to the anomalous monist, your pain will be identical to a neural event (N^1), and my pain will also identical to a neural event (N^2), but it need not be the case that N^1 and N^2 are instances of a single type of neural state. Together with functionalism (described in the next section), the token identity theory is often described as a kind of non-reductive physicalism, for unlike standard versions of the identity theory it denies that mental properties are identical to neural properties.

What account *does* the anomalous monist give of mental properties? Davidson himself gave no account of them, for he was a nominalist about properties; as he saw it, properties are not part of the fabric of reality. But realism about mental properties is

attractive, and anomalous monism's failure to give an account of them represents a sticking point for many.

Further reading

Davidson D. 1970. Mental Events. *Essays on Actions and Events*. Clarendon Press (1980).Macdonald, C. 1989. *Mind-Body Identity Theories*. Routledge.

In its modern guise, the identity theory was a reaction to behaviourism, and there are many respects in which the identity theorist's conception of the mind differs from that of the behaviourist. Analytical behaviourists assumed that the reduction of mental states to behavioural dispositions could be carried out a priori (i.e., independent of empirical investigation) and that the reductions in question were also *analytic* (i.e., followed from the meanings of the relevant terms. Identity theorists such as J. J. C. Smart (1959) rejected both of these commitments, holding that mind-brain identity statements are synthetic and a posteriori rather than analytic and a priori. Just as science has revealed that lightning is identical to electrochemical discharge and that water is identical to H_2O, so too, Smart claimed, the scientific data indicate that particular kinds of mental states are identical to particular kinds of brain states. Other identity theorists, such as Lewis (1966) and Armstrong (1968), did make substantive assumptions about the meanings of folk-psychological terms, but they too held that identifying a mental property with a neural property required scientific investigation and couldn't be established solely on the basis of philosophical analysis.

We will consider objections to the identity theory shortly, but let's first note that the identity theory qualifies as a version of physicalism for the same kind of reasons that behaviourism does. First, it satisfies the modal constraint on physicalism, for if mental states are identical to brain states, then the mental facts will supervene on the microphysical facts given that the facts about brain states presumably supervene on the microphysical facts. The identity theory also satisfies the intelligibility constraint, for if mental states are indeed identical to brain states, then the fact that the mental supervenes on the physical isn't a primitive, unanalysable fact but can instead be fully explained.[1]

The identity theory avoids some of the objections facing behaviourism, but it is vulnerable to others. For one thing, the identity theorist appears to have as much difficulty accommodating the privacy of the mental as the behaviourist does. In the same way that someone else's knowledge of my behavioural dispositions might rival (if not exceed) my own, so too someone else's knowledge of my brain states might rival (if not exceed) my own. But surely, one might think, another person's access to my mental states couldn't outstrip my access, no matter how detailed their knowledge of my brain. Even someone armed with the most accurate neuroimaging device imaginable couldn't know what it's like to be me in the way that I can.

But it is worries about the apparent contingency of the relationship between mental states and brain states that have proven most problematic for the identity theory. Writing in the 1960s and early 1970s, Hilary Putnam (1926–2016) pointed out that if pain is

identical to a certain brain state ('the firing of C-fibres', to use his physiologically implausible example), then not only will any creature with firing C-fibres be in pain, but any creature that is in pain will have firing C-fibres (Putnam 1967, 1975a). This conclusion follows from the logic of identity. (If lightning *is* electrochemical discharge, then any instance of lightning must also be an instance of electrochemical discharge and vice versa.) But, Putnam went on to suggest, it is highly improbable that there is a single type of brain state associated with all instances of pain. Instead, he argued, it is surely more plausible to suppose that pain in the octopus (for example) involves neural states and processes that are quite different from those that are implicated in human pain. In essence, Putnam's objection mirrored an objection that was instrumental in the demise of behaviourism; namely, that the behaviour associated with any particular mental state is likely to be only contingently related to it. Putnam's objection has come to be known as the *multiple realization* objection.

It is important to distinguish two versions of the multiple realization objection. One version focuses on the idea that mental properties are *actually* multiply realized. Putnam himself seems to have thought of the objection in these terms. Although it is widely assumed that mental properties are multiply realized, the issue continues to provoke debate (see Box: Multiple realization). Another version of the objection focuses on whether mental properties *could* be multiply realized. Even if pain is in fact associated with a single type of neural state in both humans and non-humans, it seems entirely possible – does it not? – that it could occur in creatures whose brains are very different from ours. Indeed, much science fiction trades on the assumption that a creature could have mental states (of the kinds we have) without having any kind of brain at all. We might call this the multiple realiz*ability* objection.

Box: Multiple realization

Although the multiple realization objection convinced a generation of philosophers to abandon the identity theory, around the turn of the century, a number of theorists began to argue that its merits had been greatly exaggerated. Bechtel and Mundale (1999), for example, pointed out that the issue of multiple realization turns crucially on exactly how states, both mental and physical, are characterized. The case for multiple realization may seem compelling if one contrasts a coarse-grained description of a mental state with a fine-grained description of a neural state. But, they argued, that would surely be illegitimate, and one should instead compare mental states with neural states that are characterized in terms of comparable levels of specificity.

Bechtel and Mundale's point is an important one, although the idea of a 'common grain' for neural and mental taxonomizing is not entirely unproblematic. In my view, however, there is enough evidence of multiple realization to put the identity theory under serious pressure even without settling issues of granularity. In fact, there is good evidence of multiple realization within our own species. The brain is highly plastic, and when an organism is deprived of input in one sensory modality the brain networks that

are associated with that modality often take on different jobs. For example, in individuals who have been blind from an early age, areas of the cortex which ordinarily process visual information are recruited for other tasks, such as processing language or the haptic-based identification of objects. These studies indicate that a single brain region or network can be implicated in different types of mental states and processes. We revisit neural plasticity in Chapter 9.

Further reading

Aizawa, K., and C. Gillett. 2009. The (Multiple) Realization of Psychological and Other Properties in the Sciences. *Mind and Language*, 24: 181–208.
Bechtel, W., and J. Mundale. 1999. Multiple Realizability Revisited: Linking Cognitive and Neural States. *Philosophy of Science*, 66(2): 175–207.
Funkhouser, E. 2007. Multiple Realizability. *Philosophy Compass*, 2 (2): 303–15.

If the multiple realizability of pain does not seem altogether convincing to you, consider mental phenomena of other kinds. Mind-brain identity theorists focus on sensations, but if the theory is to provide us with a comprehensive account of the mind, then it must also be applicable to mental phenomena in general. Consider the thought PLATO WAS BORN IN ATHENS. Not only does it seem possible that instances of this type of thought could be associated with brain states of very different kinds, but it also seems possible that a creature made entirely of silicon-based transistors could think PLATO WAS BORN IN ATHENS. But if that is so, then thinking PLATO WAS BORN IN ATHENS cannot be identified with being in a certain kind of neural state.

Building on Putnam's argument, in the mid-1970s Jerry Fodor identified a further challenge to the identity theory. Fodor approached the question of how mental states are related to brain states in terms of a wider debate about the unity of science. How are the special sciences – that is, sciences that apply only to particular domains, such as geology, biology, and psychology – related to each other and to fundamental physics? Fodor argued that it is vanishingly rare for the properties posited by one scientific theory to be identified with the properties posited by another, and that the unity of science doesn't require any such identification. Instead, a unified conception of reality requires only that we can 'explicate the physical mechanisms whereby events conform to the laws of the special sciences' (Fodor 1974: 107). Fodor did not rule out the possibility that certain types of mental properties *could* turn out to be identical to neural properties, but he made a persuasive case for thinking that we shouldn't expect such identifications to be common, and that the physicalist needn't choose between identifying mental phenomena with behavioural dispositions (as behaviourists do) and identifying them with brain states (as identity theorists do).

3.3 Functionalism

The work of Putnam and Fodor didn't just undermine the identity theory, it also provided an alternative to it: functionalism. At its heart, functionalism is the view that mental properties

are functional properties, akin to the property of being a camshaft. Something is a camshaft in virtue of its functional profile (i.e., in virtue of what it does or can do). In order to be a camshaft, an object must be able to convert rotational motion to reciprocal motion. What gives a camshaft its identity is not the substance from which it is made – one can fashion a camshaft from many kinds of materials – but the role that it plays within the system in which it is embedded. Similarly, functionalists claim that the identity of a mental state is determined by the role that it plays within the cognitive system (i.e., the mind) in which it is located.[2]

There are various views within functionalism as to how to analyse the notion of a state's functional role. However, the most influential versions of functionalism equate a state's functional role with its causal profile – that is, with the range of states that cause it (its inputs) and the range of states that it in turn causes (its output).[3] Importantly, functionalists hold that both the inputs and the outputs of a mental state invariably include other mental states. For example, pain can be triggered not only by bodily events (such as nerve damage) but also by mental events (such as memories of traumatic events). Similarly, pain has both bodily effects (increased heart rate) and mental effects (forming an intention to buy pain medication). In this respect, then, functionalism embraces a holistic conception of mentality: the identity of a particular kind of mental state is determined by how it is related to other kinds of mental states.[4]

How is functionalism related to physicalism? Functionalism is often described as a kind of non-reductive physicalism, so named because the functionalist doesn't identify mental properties with neural properties (unlike the identity theorist). Instead, functionalists think of the connection between the physical and the mental as mediated by the relation of realization: physical states necessitate mental states because they *realize* them. We can say that a property *F* realizes a property *G* if and only if: (a) *G* is identical to the property of having some property that has a certain functional role; and (b) *F* is a property that plays that functional role (Melnyk 2003). The idea is that a mental property is the second-order property of having some state or other that plays a particular kind of functional role. For example, being in pain will be identified with the second-order property of having some property that plays the pain role (e.g., being triggered by noxious stimuli and in turn triggering pain avoidance behaviour). In principle, this second-order property could be realized by different kind of physical states, for the pain role might be realized by different kinds of physical states in creatures of different kinds. Indeed, functionalists can even allow that the relevant functional roles could – in principle at least – be realized by non-physical properties.

Functionalism in the philosophy of mind is a very broad church, and there are many forms of functionalism. We don't have the space to consider every distinction within the functionalist family, but there are two contrasts that are too important to be left to one side.

The first contrast concerns how the functional roles in question are characterized. Here, the basic distinction is between long-arm functionalists and short-arm functionalists. Long-arm functionalists characterize the relevant functional roles in terms of objects and properties in the world. For example, a long-arm functionalist might distinguish a state that is caused by the perception of elm trees from a state that is caused by the perception of oak trees, even if there is no brain-based difference between these states. In this sense, then,

the long-arm functionalist holds that the external world can play a crucial role in fixing the identity of mental states. The short-arm functionalist, by contrast, restricts her attention to causal roles that are internal to the agent. From the perspective of the short-arm functionalist, differences in input and output that are described in terms of features of the environment are irrelevant. Chapter 7 explores in more detail the question of exactly how mental states depend on relations to the environment.

A second important contrast concerns how the functional roles that are essential to the identity of mental states are to be identified. Some theorists hold that these roles can be identified a priori (that is, independent of scientific investigation) because information about them is implicit in our folk theory of the mind. Here is how David Lewis puts this view:

> Think of common-sense psychology [i.e. folk psychology] as a term-introducing scientific theory, though one invented long before there was any such institution as professional science. Collect all the platitudes you can think of regarding the causal relations of mental states, sensory stimuli, and motor responses. . . . Include only platitudes which are common knowledge among us – everyone knows them, everyone knows that everyone else knows them, and so on. For the meanings of our words are common knowledge, and I am going to claim that the names of mental states derive their meaning from these platitudes.
>
> (Lewis 1972: 212)

This version of functionalism is commonly known as 'analytic functionalism', for as Lewis's comments indicate, its advocates often hold that information regarding the functional roles which determine the identities of mental states is derived from the meaning of psychological terms. However, the commitment to analyticity is not essential to this position, and I will refer to it as 'a priori functionalism' in virtue of the fact that its central commitment concerns the a priori availability of the relevant functional roles.

A very different approach to this issue is taken by a view that I will call 'a posteriori functionalism'. The a posteriori functionalist holds that the functional roles which are essential to mental states are not implicit in folk psychology but can be identified only on the basis of scientific investigation. This version of functionalism is often referred to as 'psychofunctionalism', for its advocates typically look to psychology in order to identify the functional roles of mental states. We return to the contrast between a priori functionalism and a posteriori functionalism at the end of this chapter.

Functionalism is widely regarded as a promising account of many kinds of mental states, most notably propositional attitudes such as belief, desire, and intention. Here, it is usually paired with the representational theory of mind, which holds that mental states are relations to internal representations. (We consider various aspects of the representational theory of mind in Chapters 5 and 6.) However, functionalism is significantly more controversial as an account of phenomenal states. One reason for this is that it is these states that pose the most serious privacy-related challenges, and functionalists struggle to account for the privacy of the mind in much the way that behaviourists and identity theorists do. But there is also a second reason why phenomenal states pose a challenge to functionalism – a reason that is unique to functionalism. Functionalism treats mental states as relational: what it is for a

mental to be a state of a particular kind is determined by the relations that it bears to perceptual input, behavioural output, and other mental states. Phenomenal states, by contrast, are widely taken to be non-relational (or 'intrinsic', as it is often put). What makes something a pain (for example) seems to be a matter of its particular subjective quality (its 'experiential feel'), and that quality is often taken to be independent of pain's functional profile. We return to this issue in Chapter 9. Our aim in this section has not been to evaluate functionalism, but to explore its central features, and to recognize that it provides physicalists with another conception of how mental phenomena could be nothing over and above physical phenomena (see Box: Constructing an artificial mind).

Box: Constructing an artificial mind

We can highlight the contrast between behaviourism, the identity theory, and functionalism by reflecting on how each account views the task of constructing an artificial mind. From the perspective of behaviourism, building an artificial mind requires constructing an agent with the right set of behavioural dispositions. The physical constitution and causal structure of that agent is relevant to its status as a minded entity only to the extent that it has an impact on the behaviour that the system is disposed to produce. From the perspective of the identity theory, the key to creating an artificial mind is the creation of an 'artificial brain'. What exactly that requires, however, is something of an open question, for brain states can be described at various levels of abstraction, and it is unclear precisely which levels of abstraction the identity theorist will regard as central to mentality. The functionalist will, of course, focus on issues of causal structure, arguing that to construct an artificial mind one need only duplicate the functional roles that characterize our mental states. A key question for the functionalist concerns the degree to which a system's causal structure could differ from that of the human mind and yet still count as supporting mentality. Reflection on the diversity of non-human minds suggests that the functionalist ought to adopt a relatively liberal approach to this issue, although it is unclear just how liberal the functionalist ought to be here.

3.4 Eliminativism

Behaviourism, the identity theory, and functionalism differ from each other in important ways, but they all take for granted the reality of folk-psychological states. A very different tradition within physicalism is known as 'eliminativism', a view that is so named because its advocates reject the very existence of some of the central posits of folk psychology. In the same way that talk of unicorns, fairies, and the phlogiston of 19th-century chemistry simply

reply, see Odegaard et al. [2017]). In favour of this possibility is the fact that activity in frontal areas is significantly reduced when subjects are no longer required to report their experiences (e.g., Frässle et al. 2014). Resolving this debate requires better measures of consciousness than those which we currently possess.

Second, even if 'front-of-the-brain' activity is implicated in consciousness, it is a further question whether this activity reflects the monitoring of first-order mental states. Functionalists, for example, might argue that front-of-the-brain activity is required for consciousness only because it enables the contents of first-order states to become available to the various systems that govern thought and action, and not because it is implicated in representing those states. We explore functionalist accounts of consciousness, together with some of their first-order rivals, in the following section.

9.4 Neural theories versus functional theories

The two central first-order approaches to consciousness are neural approaches and functional approaches. Both approaches recognize the importance of neural activity in generating consciousness, but they provide very different accounts of why neural activity is important. Neural theorists focus on its distinctively neural properties, whereas functionalists hold that what matters with respect to brain activity aren't its distinctively neural features but its functional profile: what it is caused by and what it in turn causes. Each approach can be developed as an account of state consciousness and phenomenal character, or its scope can be restricted to only one of these two aspects of consciousness.

Let's begin by considering neural accounts. What reasons are there to think that consciousness is either identical to (or best explained in terms of) neural activity?[10]

The mind-brain identity theorists of the 1950s were motivated by the thought that by identifying consciousness with brain activity one could avoid both the 'spooky' metaphysics of dualism on the one hand and the implausible dispositionalism of behaviourism on the other (see, e.g., Place 1956; Feigl 1958; Smart 1959). In the current context, however, this motivation is unpersuasive, for both monitoring and functional accounts also provide alternatives to dualism and behaviourism. What's needed here is some reason to think that consciousness is grounded in (or even just correlated with) the brain's neural properties as opposed to its representational or functional ones.

Neural theorists sometimes appeal to the effects of neural interventions on consciousness as evidence for their view. Some neural interventions, such as those caused by general anaesthetics, eliminate consciousness entirely; others, however, have an impact only on specific types of experiences (e.g., Fox et al. 2020; Winawer & Parvizi 2016). For example, lesions to an area in the visual cortex called MT+ can cause akinetopsia, a deficit in the ability to see motion, whereas stimulating MT+ with transcranial magnetic stimulation (TMS) can trigger experiences of motion. Does the fact that it is possible to intervene on consciousness in these ways show that consciousness is best understood in neural terms?

No. In fact, there are multiple problems with the interventionist argument. First, the evidence for correlations between specific types of brain activity and specific conscious experiences is rather weak. Although the brain does contain areas and networks that care

about certain kinds of stimuli more than others, there appear to be few one-to-one mappings from particular brain areas and networks onto particular forms of experience, and even relatively specialized areas and networks are implicated in many types of conscious state (Anderson 2014). To take just one example of this phenomenon, consider the so-called 'pain matrix', a network involving the primary and secondary somatosensory cortices, the insula, and the anterior cingulate cortex, which is typically active when people experience pain. Although the pain matrix is correlated with pain, we cannot map activity in the pain matrix onto pain, for the pain matrix is also activated by stimuli that are highly salient (i.e., attention-grabbing) but not painful (Iannetti & Mouraux 2010; Mouraux et al. 2011). Neural plasticity – the fact that brain areas that are typically used to process stimuli of one kind are, in some cases, used to process stimuli of another kind – provides a closely-related challenge to the interventionist argument (Box: Cortical dominance and cortical deference).

Box: Cortical dominance and cortical deference

What happens when the connections between the brain and the world are rewired, so that brain areas that normally receive one type of input (say, visual input) receive a novel type of input (say, auditory input)? It depends. In some cases activity in the target area leads to the kinds of experiences that are normally associated with it, but in other cases it leads to experiences associated with the new type of input. Hurley and Noë (2003) apply the label 'cortical dominance' to cases of the first type (because the nature of the experience is in line with that which is typically associated with the cortical region in question), and they apply the label 'cortical deference' to cases of the second type (because consciousness defers to the nature of the input).

Cortical dominance is illustrated by the phantom limb phenomenon. Normally, tactile input from the arm and face maps onto adjacent cortical areas. However, after amputation of part of the arm, tactile input from the face invades deafferented cortex whose normal conscious expression involves feeling one's arm being touched. When this area of cortex is activated by input from the face, it continues to retain this conscious expression. So, when the face of someone with a phantom limb is stroked, they will often experience their (missing) limb being stroked (in addition to feeling their face being stroked (Ramachandran & Hirstein 1998). Cortical deference is illustrated by neural activity in blind children, in which areas of the cortex that are ordinarily dedicated to vision are used to process speech and haptic information (Bedney et al. 2015; Hamilton & Pascual-Leone 1998).

Cortical dominance is unsurprising from the perspective of neural theories, but such theories struggle to explain cortical deference. Why should modulating the *input* to an area of cortex change its conscious expression if consciousness is to be accounted for

in neural terms? By contrast, other approaches to consciousness, such as representationalism (see Section 9.5), have no difficulty accounting for cortical deference but they struggle to explain cortical dominance.

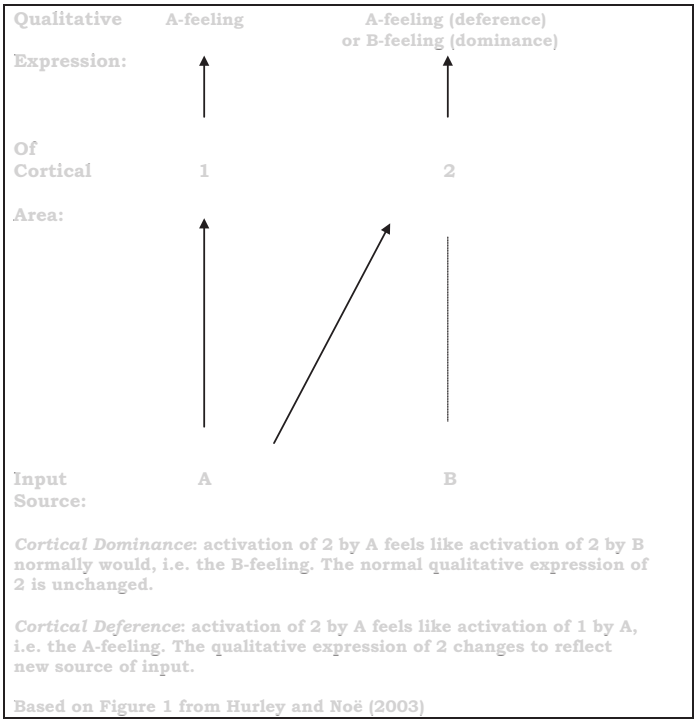

Figure 9.2 Cortical deference versus cortical dominance

Source: Reproduced from Susan Hurley and Alva Noë (2003), 'Neural Plasticity and Consciousness', *Biology and Philosophy*, 18: 131–68, with the kind permission of Alva Noë. Figure by Miriam Dym, Susan Hurley and Alva Noë.

Further reading

Hurley, S., and A. Noë. 2003. Neural Plasticity and Consciousness. *Biology and Philosophy*, 18(1): 131–68.

The second problem with the interventionist argument is that other accounts of consciousness can also explain why certain interventions on brain activity have particular effects on consciousness. Monitoring theorists can account for such effects by arguing that neural interventions alter the capacity to monitor first-order mental states or by modulating the first-order

mental states themselves, and functionalists can argue that neural interventions affect consciousness by modulating the functional profile of the relevant mental states. In sum: not only is it unclear whether there are systematic relations between neural activity and consciousness, to the extent that systematicity obtains neural accounts may not provide the best explanation of it.

Let's turn now to criticisms of the neural approach. Perhaps the most influential criticism is the objection from chauvinism: because neural theories entail that consciousness is limited to entities with neural properties, they seem to be unreasonably conservative. Even if human consciousness is best explained in neural terms, is it really plausible to suppose that all possible forms of consciousness require neural activity? Couldn't a suitably constructed AI system be conscious, or at least pass all of our best tests for consciousness?

In response to this objection some theorists bite the bullet, arguing that it is simply begging the question against neural theories of consciousness to assume that non-neural systems can be conscious. In a sense they are right: the objection *does* beg the question. But that doesn't render it irrelevant or unfair. The point is that there could be systems which are unconscious (by the lights of the neural theory) but which would pass all of our current tests for consciousness. (Indeed, such systems might also pass any future test for consciousness!) It doesn't, of course, follow that such systems would be conscious, but surely there would be something peculiar in recognizing that a system has passed our best tests for consciousness but refusing to regard it as conscious.

Not only does the objection from chauvinism undermine the neural approach to consciousness, it also points towards another view: functionalism. Functionalists argue that consciousness is best understood in terms of the functional profile of brain states and processes, where a state's functional profile is typically understood in terms of its causal role: what causes it and what it in turn causes. As we noted in Chapter 3, there are many forms of functionalism.

One important contrast is between a priori functionalism (also known as analytic functionalism) and a posteriori functionalism (also known as psychofunctionalism). A priori functionalists hold that the functional roles associated with consciousness can be identified via conceptual analysis, whereas a posteriori functionalists hold that the functional roles associated with consciousness can be identified only by empirical research in (for example) psychology and neuroscience. Although a priori functionalism has been very influential as an account of certain kinds of mental states (e.g., belief, desire, and intention), it has had significantly less influence as an account of consciousness. The problem, in a nutshell, is that the central concept of consciousness, PHENOMENAL CONSCIOUSNESS, doesn't seem to be a functional concept. Phenomenal consciousness itself may turn out to be a functional property, but the concept of phenomenal consciousness does not present it as such.

For the reasons just given, most functionalist accounts of consciousness are versions of a posteriori functionalism. For example, Michael Tye's (1995) PANIC account holds that representations are conscious when they are poised for the flexible control of thought and action. Jesse Prinz's (2012) attended intermediate representations (AIR) theory is also a version of a posteriori functionalism, for it treats attention – understood in terms of access to working memory – as a key factor in explaining why certain mental states are conscious. But perhaps the most influential version of a posteriori functionalism is Bernard Baars's (1988) Global Workspace Theory (GWT). According to the GWT, representations are conscious when they

enter a 'global workspace', which makes their contents available to a wide array of 'consuming systems', such as working memory, metacognition, and attention.[11]

GWT is a version of what I will call 'quasi-functionalism', for it aims to explain only why conscious mental states are conscious, and says nothing about phenomenal character. Other functionalist accounts are comprehensive, and invoke functional roles to explain not only why some mental states are conscious but also why different kinds of conscious states have different phenomenal characters. Quasi-functionalism is the more influential of the two views, but comprehensive functionalism does have its advocates (e.g. Shoemaker 1975).

So much for the differences between functional accounts. What should we make of the approach in general?

One of the leading motivations for functionalism is the fact that conscious states seem to have a functional profile that distinguishes them from unconscious states. The contents of consciousness are, in general at least, available for the direct control of thought and action. They are reportable, they guide our voluntary behaviour, and they inform our deliberations. By contrast, although the contents of unconscious states can have indirect effects on thought and action, they aren't available for the direct control of thought and action in the ways in which the contents of conscious states are. Let's call this claim the *availability thesis*. According to the *availability argument*, the best explanation for the truth of the availability thesis is that consciousness is to be explained in terms of functional properties.

Let's consider three objections to the availability argument. The first concerns the status of the availability thesis. In effect – the objection runs – we presuppose the availability thesis in deciding whether or not a mental state is conscious, for states whose contents are not generally available for the control of thought and action simply aren't regarded as conscious. In other words, it's possible that we don't see counterexamples to the availability thesis not because they don't exist but simply because our methods for detecting consciousness are blind to their possibility. Were we to adopt other criteria for the ascription of consciousness then the availability thesis might not appear to be plausible.

A second issue concerns the scope of the availability thesis. What kinds of agents does it apply to? Neurotypical adult human beings? All human beings? All mammals? All biological organisms? All agents whatsoever, irrespective of their nature? The most ambitious versions of functionalism take the view to apply to conscious agents in general. But the evidence that we have for the availability thesis (to the extent that we have any at all) concerns only human beings, a very small sample of possible conscious agents. Any attempt to apply functionalism more broadly will need to justify the assumption that the functional role associated with consciousness in humans is indicative of its functional role in systems of other kinds.

A final problem with the availability argument concerns the inference from the availability thesis to the truth of functionalism. In effect, the availability argument holds that because conscious states have a distinctive functional profile consciousness itself is to be explained or understood in terms of functional properties. But, one might argue, if consciousness is to be understood in functional terms, then one can't *explain* why conscious states have a distinctive functional profile, for that would be circular. In effect, one would be claiming that conscious states have a distinctive functional profile because they have a distinctive functional profile – true but vacuous! Instead, the objection goes, only *non-functional* theories can *explain* why conscious states have a distinctive functional profile.

Functionalism's critics don't merely take issue with the arguments for functionalism; they also provide objections to the view. One of the most influential of these is Ned Block's (1978) China-Brain argument. Block asks us to imagines a scenario in which the residents of China simulate for an hour the functional roles of the neurons in a normal human brain by communicating with each other via walkie-talkies. This activity creates a massively distributed agent: 'China-Brain'. The only difference between you and China-Brain is the medium in which the relevant functional states are realized: roughly speaking, your functional states are realized by the interactions between neurons, whereas China-Brain's functional states are realized by the interactions between Chinese citizens. For the functionalist, however, all that matters when it comes to consciousness are a system's functional properties. Thus (Block argues), the functionalist should conclude that China-Brain has the conscious states that you do. But, Block claims, it is clearly absurd to suppose that China-Brain would have conscious states of any kind, let alone the kinds of conscious states that you enjoy. In effect, Block argues that functionalists are committed to an overly liberal view of the distribution of consciousness.

There are two lines of response open to the functionalist. The first denies that the functional roles relevant to consciousness can be captured by simply duplicating the causal relations between neurons. According to 'long-arm' functionalists, the input and output states that determine consciousness involve states of the environment rather than states that are internal to the cognitive system. In this version of functionalism, China-Brain would need to interact with its environment in roughly the ways that we do in order to share our conscious states.

Other functionalists – 'short-arm' functionalists – accept that China-Brain could have the kinds of functional states on which consciousness supervenes, but they deny that it is 'clearly absurd' to ascribe consciousness to China-Brain. After all (they point out), China-Brain will share our behavioural capacities: it will produce introspective reports; it will engage in voluntary behaviour; and it will monitor its own mental states. If asked, it would be able to tell you what it's like to taste durian, how pains differ from itches, and whether it prefers Aretha Franklin to Queen. China-Brain may not in fact be conscious, but it does seem likely that we would be tempted to view it as conscious. And if that's right, then the China-Brain argument leaves functionalism unscathed.

9.5 Intentionalism

Let's bring this chapter to a close by considering an account of consciousness that focuses exclusively on the explanation of phenomenal (or experiential) character. The account in question is often known as 'representationalism', but I prefer the less common label 'intentionalism'. We encountered a restricted form of intentionalism in Chapter 4, when we considered the claim that the phenomenal character of perception is fixed by its intentional content. On this view, the phenomenal differences between tasting coffee, hearing the opening notes of Miles Davis's *Kind of Blue*, and looking at a stand of redwoods can be understood in terms of the intentional properties of these experiences. Here, our focus is on intentionalism understood as a claim about *all* forms of phenomenal character. Thus understood, the claim is that the phenomenal feel of even bodily sensations and emotions can be accounted for in terms of intentionality.